RATIO & PROPORTION

LESSON-1: INTRODUCTION

RATIO = Comparision of 2 or more items. After comparision, that comparision value will be represented in sort of a numbers.

Eg: $2:1 = \frac{2}{1}$

PROPORTION = Equality of two ratios

$::$ ⇒ These four dots is proportion, these dots mean equal to

i.e. $a:b :: c:d$

(or)

$a:b = c:d$

TYPES OF RATIO'S

1. Duplicate Ratio → just square the term

Eg: what is duplicate ratio of 3:7

$3^2:7^2 = 9:49$

2. Triplicate Ratio → just cube the term

Eg: $3:7 \rightarrow 3^3:7^3 = 27:343$

3. Sub-Duplicate Ratio → take square root of given term

Eg: $9:25 \rightarrow \sqrt{9}:\sqrt{25} = 3:5$

4. Sub-Triplicate Ratio → take cube root of given term

Eg: $125:343 \rightarrow \sqrt[3]{125}:\sqrt[3]{343} = 5:7$

5. Inverse ratio → just reverse the terms

Eg: $5:7 \rightarrow 7:5$

6. Compound ratio → $\frac{\text{multiply all the 1st terms}}{\text{multiply all the 2nd terms}}$

Eg: $\begin{matrix} a:b & c:d & e:f \\ 1:3 & 2:9 & 4:7 \end{matrix} \rightarrow \frac{a \times c \times e}{b \times d \times f} = \frac{1 \times 2 \times 4}{3 \times 9 \times 7}$

* RATIO SHOULDN'T BE IN FRACTION

Sometimes in question itself, they will give the ratio in fraction, In this case what you have to do is first take the LCM of the numbers & convert the given value in terms of integer & then start solving the question.

Eg: $a:b = \frac{2}{9} : \frac{1}{3}$

$= \frac{2 : 3}{9}$

$\therefore\ a:b = 2:3$

In case you get final answer in fraction, do the same thing like above.

LESSON-2: FINDING RATIO'S

* Let $a : b = 2 : 3$

$\boxed{\begin{matrix} a = 2 \\ b = 3 \end{matrix}}$ $\qquad$ $\boxed{\begin{matrix} a = 2x \\ b = 3x \end{matrix}}$

difference b/w these 2

In given question if the values are in ratio & final answer what they are asking is also in ratio then use this logic

In given question if the values are in ratio & find answer what they are asking is in numbers then use this logic

Q) If $A:B:C = 2:3:4$, then what is the value of $A/B : B/C : C/A$?

Sol: $\underbrace{A:B:C = 2:3:4}_{\text{question given in ratio}}$ $\underbrace{\frac{A}{B} : \frac{B}{C} : \frac{C}{A}}_{\text{answer also asking in ratio}}$

So, you can directly substitute the values

$$\frac{A}{B} : \frac{B}{C} : \frac{C}{A} = \frac{2}{3} : \frac{3}{4} : \frac{4}{2}$$ → ratio's are in fraction, So this is not final answer.

Take LCM

$$\frac{8 : 9 : 24}{12}$$

Answer = $8:9:24$

Q) If $A:B = 2:3$ and $B:C = 4:5$, then $A:B:C$ is:

Sol: $A:B = 2:3$ $B:C = 4:5$ $A:B:C = ?$

$$\begin{array}{ccccc} A & : & B & : & C \\ 2 & : & 3 & : & \boxed{3} \\ \boxed{4} & : & 4 & : & 5 \\ \hline \underset{(2\times4)}{8} & : & \underset{(3\times4)}{12} & : & \underset{(3\times5)}{15} \end{array}$$

→ write the number nearer to this □

write number nearer to this □ →

→ by multiplying terms

Answer: $8:12:15$

* There will be many technique's to solve this type of question. So learn any one technique only.

Q) If $A:B = \frac{1}{2}:\frac{1}{3}$ and $B:C = \frac{1}{2}:\frac{1}{3}$, then $A:B:C = ?$

Sol: $A:B = \frac{1}{2}:\frac{1}{3} = \frac{3:2}{6} \Rightarrow A:B = 3:2$

$B:C = \frac{1}{2}:\frac{1}{3} = \frac{3:2}{6} \Rightarrow B:C = 3:2$

$$\begin{array}{ccccc} A & : & B & : & C \\ 3 & : & 2 & : & \boxed{2} \\ \boxed{3} & : & 3 & : & 2 \\ \hline 9 & : & 6 & : & 4 \end{array}$$

Ans = $9:6:4$

Q) If $a:b = 5:7$ and $c:d = 2a:3b$, then $ac:bd$ is

Sol: $a:b = 5:7$ $\qquad c:d = 2a:3b$

$\frac{a}{b} = \frac{5}{7}$ $\qquad \frac{c}{d} = \frac{2a}{3b}$

$$ac:bd = \frac{ac}{bd} = \frac{5 \times 2a}{7 \times 3b} = \frac{5}{7} \times \frac{2}{3} \times \frac{a}{b}$$

$$= \frac{5}{7} \times \frac{2}{3} \times \frac{5}{7}$$

$$= \frac{50}{147}$$

Q) If $3A = 5B$ and $4B = 6C$, then $A:C = ?$

Sol: $\frac{A}{B} = \frac{5}{3}$ $\qquad \frac{B}{C} = \frac{6}{4}$

$A:B = 5:3$ $\qquad B:C = 6:4$

$$\begin{array}{ccccc} A & : & B & : & C \\ 5 & : & 3 & : & \boxed{3} \\ \boxed{6} & : & 6 & : & 4 \\ \hline 30 & : & 18 & : & 12 \end{array}$$

$\therefore A:B:C = 30:18:12$

$A:C = \cancel{30}^{5} : \cancel{12}^{2}$

$\boxed{A:C = 5:2}$

Q) If $a:5 = b:7 = c:8$ then $(a+b+c)/a$ is equal to

Sol: $a:5 = b:7 = c:8$ → This is a proportion because 2 ratio's are equated

$$\underbrace{\frac{a}{5} = \frac{b}{7} = \frac{c}{8}}_{\text{given in ratio}}$$

$$\underbrace{\frac{a+b+c}{a}}$$

asking answer in numbers i.e $(a+b+c)$ divide by a. so take in terms of x

i.e. $a = 5x$
$b = 7x$
$c = 8x$

$$\therefore \frac{a+b+c}{a} = \frac{5x+7x+8x}{5x} = \frac{\cancel{20x}^{4}}{\cancel{5x}} = 4$$

Q) If $(x/y) = (6/5)$, find the value $(x^2+y^2)/(x^2-y^2)$

Sol: $\frac{x}{y} = \frac{6}{5} \Rightarrow x = 6a \quad y = 5a$

$$\frac{x^2+y^2}{x^2-y^2} = \frac{(6a)^2+(5a)^2}{(6a)^2-(5a)^2} = \frac{36a^2+25a^2}{36a^2-25a^2} = \frac{61\cancel{a^2}}{11\cancel{a^2}}$$

$$= \frac{61}{11}$$

Q) If $x:y = 8:9$, then $5x-4y : 3x+2y$ is equal to:

Sol: $x:y = 8:9$

$$\underbrace{5x-4y : 3x+2y}$$

↓

asking answer in ratio, but before that we need to some calculations

So, $x = 8a$
$y = 9a$

$$\frac{5x-4y}{3x+2y} = \frac{5(8a)-4(9a)}{3(8a)+2(9a)} = \frac{40a-36a}{24a+18a} = \frac{4\cancel{a}}{42\cancel{a}}$$

$= 2:21$

Q) If $x/2y = 6/7$, the value of $(x-y)/(x+y) + 14/19$

Sol: $\frac{x}{2y} = \frac{6}{7} \Rightarrow \frac{x}{y} = \frac{12}{7}$

$$\frac{x-y}{x+y} + \frac{14}{19} = \frac{12a-7a}{12a+7a} + \frac{14}{19}$$

$$= \frac{5\cancel{a}}{19\cancel{a}} + \frac{14}{19} = \frac{19}{19} = 1$$

Q) If $a:b = 2:3$ and $b:c = 4:5$ is then $a^2:b^2:bc$ is

Sol: $a:b = 2:3 \qquad b:c = 4:5$

$a^2:b^2:bc = ?$

Find $a:b:c \Rightarrow$

a	b	c
2	3	$\boxed{3}$
$\boxed{4}$	4	5
8 :	12 :	15

$$a^2:b^2:bc = (8)^2 : (12)^2 : (12)(15)$$

$$= 64 : 144 : 180$$

$$= 16 : 36 : 45$$

Q) If $A:B = 2:3$, $B:C = 4:5$ and $C:D = 6:7$, what is the value of $A:B:C:D$?

Sol:

A	B	C	D
2	3	$\boxed{3}$	$\boxed{3}$
$\boxed{4}$	4	5	$\boxed{5}$
$\boxed{6}$	$\boxed{6}$	6	7
$\cancel{48}$:	$\cancel{72}$:	$\cancel{90}$:	$\cancel{105}$
16	24	30	35

$\therefore A:B:C:D = 16:24:30:35$

Q) If a:b = 2/9 : 1/3, b:c = 2/7 : 5/14, d:c = 7/10 : 3/5, then find a:b:c:d

Sol: a:b $\frac{a}{b} = \frac{2}{9} : \frac{1}{3} = \frac{2:3}{9} \Rightarrow a:b = 2:3$

$b:c = \frac{b}{c} = \frac{2}{7} : \frac{5}{14} = \frac{4:5}{14} \Rightarrow b:c = 4:5$

$d:c = \frac{7}{10} : \frac{3}{5} = \frac{7:6}{10} \Rightarrow d:c = 7:6$

a	b	c	d
2	3	$\boxed{3}$	$\boxed{3}$
$\boxed{4}$	4	5	$\boxed{5}$
$\boxed{6}$	$\boxed{6}$	6	7
$\cancel{48}$ 16	$\cancel{72}$ 24	$\cancel{96}$ 30	$\cancel{105}$ 35

= 16:24:30:35

LESSON-3: DIVIDED INTO PARTS

Q) If Rs. 1000 is divided between A and B in the ratio 3:2, then A will receive

Sol: Total amount = 1000 ⟶ A : B
3 : 2

$A = \frac{3}{5_{(3+2)}} \times \text{Total} = \frac{3}{\cancel{5}} \times \cancel{1000}^{200}$

A = 600//

Q) If 78 is divided into three Parts which are in the ratio 1 : 1/3 : 1/6 the middle part is

Sol: $a:b:c = 1 : \frac{1}{3} : \frac{1}{6}$

Convert ratio into integer

$$a : b : c = \frac{6 : 2 : 1}{6} = 6 : 2 : 1$$

$$\text{middle part} = \frac{2}{6+2+1} \times 78$$

$$= \frac{2}{\cancel{9}_{3}} \times \cancel{78}^{26}$$

$$= \frac{52}{3}$$

9) Rs. 33,630 are divided among A, B and C in such a manner that the ratio of the amount of A to that of B is 3:7 and the ratio of the amount of B to that of C is 6:5. The amount of money received by B is

Sol: $A : B = 3 : 7$

$B : C = 6 : 5$

A	:	B	:	C
3	:	7	:	$\boxed{7}$
$\boxed{6}$	:	6	:	5
18	:	42	:	35

$$B = \frac{42}{18+42+35} \times 33,630 = \frac{42}{\cancel{95}_{19}} \times \cancel{33,630}^{\cancel{6726}^{354}}$$

$$B = 42 \times 354$$

$$= 14,868$$

9) Rs. 3400 is divided among A, B, C, D in such a way that the share of A and B, B and C, C and D may be as 2:3, 4:3 and 2:3. The sum of shares of B and D is.

Sol: $A : B = 2 : 3$ $\quad B : C = 4 : 3$ $\quad C : D = 2 : 3$

A	B	C	D
2	3	$\boxed{3}$	$\boxed{3}$
$\boxed{4}$	4	3	$\boxed{3}$
$\boxed{2}$	$\boxed{2}$	2	3

$(2\times4\times2) : (3\times4\times2) : (3\times3\times2) : (3\times3\times3)$

$$A : B : C : D = 16 : 24 : 18 : 27$$

Sum of shares of B and D $= \dfrac{24+27}{16+24+18+27} \times 3400$

$= \dfrac{51}{\cancel{85}} \times \overset{40}{\cancel{3400}} = 2040$

8) By mistake instead of dividing Rs. 117 among A, B and the ratio $1/2 : 1/3 : 1/4$ it was divided in the ratio of $2:3:4$. Who gains the most and by how much?

Sol:

117

$\frac{1}{2} : \frac{1}{3} : \frac{1}{4}$

$= \dfrac{6:4:3}{12}$

$= \boxed{6:4:3}$

$\boxed{2:3:4}$

(divided wrong)

Add all 3 persons amount $\Rightarrow 6x + 4x + 3x = 117$

$13x = 117$

$x = 9$

A share $= 6x = 6(9) = 54$

B share $= 4x = 4(9) = 36$

C share $= 3x = 3(9) = 27$

<u>In wrongly divided case:</u>

Add all 3 persons amount $\Rightarrow 2x + 3x + 4x = 117$

$9x = 117$

$x = 13$

A share $= 2x = 2(13) = 26$

B share $= 3x = 3(13) = 39$

C share $= 4x = 4(13) = 52$

Actual calculations | Wrong Calculations

$A = 54 \longrightarrow A = 26$

$B = 36 \xrightarrow{+3} B = 39$

$C = 27 \xrightarrow{+25} C = 52$

The person who gains most is C by 25 because of wrong calculations.

Q) If a sum of money is to be divided among A, B, C such that A's share is equal to twice B's share and B's share is 4 times C's share, then their shares are in the ratio:

Sol: $A = 2B$ $\qquad$ $B = 4C \Rightarrow \frac{B}{C} = \frac{4}{1}$

$\frac{A}{B} = \frac{2}{1} \Rightarrow A:B = 2:1$ $\qquad$ $B:C = 4:1$

$A : B : C$

$2 : 1 : \boxed{1}$

$\boxed{4} : 4 : 1$

$8 : 4 : 1$

Q) Divide Rs 1250 among A, B, C so that A gets 2/9 of B's share and C gets 3/4 of A's share. Find the shares of A, B and C.

Sol: $A = \frac{2}{9}B$ $\qquad$ $C = \frac{3}{4}A$

$\frac{A}{B} = \frac{2}{9}$ $\qquad$ $\frac{C}{A} = \frac{3}{4}$

$A : B = 2:9$ $\qquad$ $C : A = 3:4$

$$A : B : C$$
$$2 : 9 : \boxed{9}$$
$$3 : \quad : 4$$

Here, Since $C = \frac{3}{4}A$

$$C = \frac{\cancel{3}}{\cancel{4}_2}\left(\frac{\cancel{2}}{\cancel{9}_3}B\right) \Rightarrow C = \frac{1}{6}B$$

$$C : B = 1 : 6$$

$$A : B : C$$
$$2 : 9 : \boxed{9}$$
$$\boxed{6} : 6 : 1$$

$$(2\times\cancel{6}_2) : (9\times\cancel{6}_2) : (\cancel{9}_3\times 1)$$
$$4 : 18 : 3$$

$\therefore A:B:C = 4:18:3$

Total sum of $A, B, C = 4x + 18x + 3x$

$$25x = 1250$$
$$\boxed{x = 50}$$

A share $= 4x = 4(50) = 200$

B share $= 18x = 18(50) = 900$

C share $= 3x = 3(50) = 150$

Q) Rs. 2010 are to be divided among A, B, C in such a way that if A gets Rs. 5, then B must get Rs. 12 and if B gets Rs. 4, then C must get Rs. 5.50. The share of C will exceed that of B by.

Sol: $A : B = 5 ; 12$

$B : C = 4 : 5.50$
$= 4 : \frac{\cancel{550}^{11}}{\cancel{100}_2} = \frac{8 : 11}{2}$

$B : C = 8 : 1$

$$A : B : C$$
$$5 : 12 : \boxed{12}$$
$$\boxed{8} : 8 : 1$$

$$5\times\cancel{8}_2 : 12\times\cancel{8}_2 : \cancel{12}_3\times 1$$
$$10 \quad 24 \quad 3$$

$A:B:C = 10:24:3$

$C - B = 33 - 24 = 9$

$$\frac{9}{\cancel{67}} \times \overset{30}{\cancel{2012}} = 270$$

LESSON-4: BASED ON NUMBERS

Q) The ratio of two numbers is 3:8 and their difference is 115. The larger number is.

Sol: $a_1 : a_2 = 3 : 8$

$3 \to 3x$ (Small), $8 \to 8x$

$a_1 \sim a_2 = 115$

$3x \sim 8x = 115$

$5x = 115$

$x = 23$

larger number $= 8x = 8(23)$

$= 184$

Q) The ratio of two numbers is 10:7 and their difference is 105. The sum of the numbers are

Sol: $a_1 : a_2 = 10 : 7$

$a_1 \sim a_2 = 105$

$10x \sim 7x = 105$

$3x = 105$

$x = 35$

sum $= 10x + 7x$

$= 10(35) + 7(35)$

$= 350 + 245$

$= 595$

Q) The sum of two numbers is 40 and difference between them is 4. Find the numbers.

Sol: $a_1 + a_2 = 40$

$a_1 - a_2 = 4$

$2a_1 = 44$

$a_1 = \frac{44}{2}$

$\boxed{a_1 = 22}$

$a_1 + a_2 = 40 \Rightarrow 22 + a_2 = 40$

$a_2 = 40 - 22$

$\boxed{a_2 = 18}$

Q) Three numbers are in the ratio 3:2:5 and the sum of their squares is 1862. What are the three numbers?

Sol: $a_1 : a_2 : a_3 = 3:2:5$

$a_1^2 + a_2^2 + a_3^2 = 1862$

$(3x)^2 + (2x)^2 + (5x)^2 = 1862$

$9x^2 + 4x^2 + 25x^2 = 1862$

$38x^2 = 1862$

$x^2 = \frac{\cancel{1862}^{49}}{\cancel{38}} \Rightarrow x = \sqrt{49}$

$\boxed{x = 7}$

Numbers $= 3x = 3(7) = 21$

$2x = 2(7) = 14$

$5x = 5(7) = 35$

Q) Of the three numbers, the ratio of the first and the second is 8:9 and that of the second and third is 3:4. If the product of the first and third number is 2400, then the second number is:

Sol: $a_1 : a_2 = 8:9$ $\quad a_2 : a_3 = 3:4$

$a_1 a_3 = 2400$

$a_1 : a_2 : a_3$

$8 : 9 : \boxed{9}$

$\boxed{3} : 3 : 4$

$8 \times \cancel{3} : 9 \times \cancel{3} : \cancel{9}^{3} \times 4$

$8 : 9 : 12$

$a_1 : a_2 : a_3 = 8 : 9 : 12$

$a_1 a_3 = \; = 2400$

$(\cancel{8}x)(\cancel{12}x) = \dfrac{\cancel{2400}}{\cancel{200}}\; 25$

$x^2 = 25$

$\boxed{x = 5}$

$a_2 = 9x = 9(5) = 45$//

Q) Three numbers are in the ratio 1/2 : 2/3 : 3/4. The difference between the greatest and the smallest number is 36. The numbers are?

Sol: $a_1 : a_2 : a_3 = \frac{1}{2} : \frac{2}{3} : \frac{3}{4} = \dfrac{6 : 8 : 9}{12}$

$= \underset{\text{small}}{6} : 8 : \underset{\text{greatest}}{9}$

$a_3 \sim a_1 \Rightarrow 9x - 6x = 36$

$3x = 36$

$\boxed{x = 12}$

Numbers are : $6x = 6(12) = 72$

$8x = 8(12) = 96$

$9x = 9(12) = 108$

Q) There is a ratio of 5 : 4 between two numbers. If 40% of the first number is 12 then what would be the 50% of the second number?

Sol: $a_1 : a_2 = 5 : 4$

40% of $a_1 = 12$

50% of $a_2 = ?$

40% of $a_1 = 12$

40% of $5x = 12$ $\Rightarrow \frac{\cancel{40}^{2}}{\cancel{100}_{2}} \times \cancel{5}x = \cancel{12}^{6}$

$\boxed{x = 6}$

50% of $a_2 = \frac{50}{100} \times 4x = \frac{\cancel{50}}{\cancel{100}_{2}} \times \cancel{4}^{2} \times 6$

$= 12$

LESSON-5: BOYS & GIRLS

Q) The ratio of the number of boys to girls of a school with 504 students is 13:11. What will be the new ratio if 12 more girls are admitted?

Sol: Initial total students = 504

Boys : girls = 13 : 11

No. of Boys $= \frac{13}{\cancel{24}} \times \cancel{504}^{21} = 21 \times 13 = 273$

No. of Girls $= \frac{11}{\cancel{24}} \times \cancel{504}^{21} = 21 \times 11 = 231$

12 more girls added

New No. of Girls $= 231 + 12 = 243$

New ratio $= \frac{\text{Boys}}{\text{Girls}} = \frac{\cancel{273}^{91}}{\cancel{243}_{81}} = 91 : 81$

Q) In classroom, Boys and Girls are in the ratio of 5:7. If 9 left from each of them, their ratio becomes 7:11. The difference of boys and Girls is:

Sol: Initial ratio of Boys and Girls = 5:7

No. of Boys = $5x$

No. of girls = $7x$

9 left from each of them

New ratio of $B:G = 7:11$

$$\frac{5x-9}{7x-9} = \frac{7}{11}$$

$11(5x-9) = 7(7x-9) \Rightarrow 55x - 99 = 49x - 63$

$55x - 49x = 99 - 63$

$6x = 36$

$\boxed{x = 6}$

No. of Boys - No. of Girls $= 7x - 5x$
$= 2x$
$= 2(6)$
$= 12$

Q) The students in three classes are in the ratio 2:3:5 If 40 students are increased in each class the ratio changes to 4:5:7 Originally the total number of students was.

Sol: Initially $C_1 : C_2 : C_3 = 2:3:5$
$2x \quad 3x \quad 5x$

40 students increased in each class

New ratio $= 4:5:7$

$2x+40 : 3x+40 : 5x+40 = 4:5:7$

Take any 2 ratio value to find x

$$\frac{2x+40}{3x+40} = \frac{4}{5} \Rightarrow 5(2x+40) = 4(3x+40)$$
$10x + 200 = 12x + 160$

$200 - 160 = 12x - 10x$

$40 = 2x$

$\boxed{x = 20}$

Original total strength $= 2x + 3x + 5x$
$= 10x$
$= 10(20)$
$= 200$

9) The ratio of boys and girls in a college is 5:3. If 50 boys leave the college and 50 girls join the college, the ratio becomes 9:7. The number of boys in the college is:

Sol: Initial boys : girls $= 5 : 3$

$5x \quad 3x$

50 boys leave, 50 girls join

New ratio $= \frac{\text{Boys}}{\text{Girls}} = \frac{5x-50}{3x+50} = \frac{9}{7}$

$$7(5x-50) = 9(3x+50)$$
$$35x - 350 = 27x + 450$$
$$35x - 27x = 450 + 350$$
$$8x = 800$$
$$\boxed{x = 100}$$

No. of boys in college $= 5x$

$= 5(100)$

$= 500$

9) The ratio of the number of boys to that of girls in a group becomes 2:1 when 15 girls leave. But afterwards when 45 boys also leave, the ratio becomes 1:5. Originally the number of girls in the group was.

Sol: Initially let Boys $= B$ Girls $= G$

15 girls leave:

$$\frac{\text{Boys}}{\text{Girls}} = \frac{B}{G-15} = \frac{2}{1}$$

$$B = 2(G-15)$$
$$B = 2G - 30$$
$$2G - B = 30 \quad \text{--- (1)}$$

45 boys leave:

$$\frac{\text{Boys}}{\text{Girls}} = \frac{B-45}{G-15} = \frac{1}{5}$$

$$5(B-45) = G-15$$
$$5B-225 = G-15$$
$$5B-G = 225-15$$
$$5B-G = 210 \quad \text{—— (2)}$$

Put (1) in (2)

$$5(2G-30) - G = 210$$
$$10G - 150 - G = 210$$
$$9G = 210+150$$
$$9G = 360$$
$$\boxed{G = 40}$$

9) The total number of students in a school was 660. The ratio between boys and girls was 13:9. After some days, 30 girls joined the school and some boys left the school and new ratio between boys and girls become 6:5. The number of boys who left the school is

Sol: Total = 660

Boys : Girls = 13:9

$$\text{No. of Boys} = \frac{13}{\cancel{22}} \times \overset{30}{\cancel{660}} = 13 \times 30 = 390$$

$$\text{No. of Girls} = \frac{9}{\cancel{22}} \times \overset{30}{\cancel{660}} = 9 \times 30 = 270$$

30 girls joined, some boys left, new ratio = 6:5

$$\frac{\text{Boys}}{\text{Girls}} = \frac{390-B}{270+30} = \frac{6}{5}$$

$$\cancel{5}(390-B) = 6(\overset{60}{\cancel{300}})$$
$$390-B = 360$$
$$B = 30$$

$\therefore$ 30 boys left the school

Q) In a college union, there are 48 students. The ratio of the number of boys to the number of girls is 5:3. The number of girls to be added in the union, so that the number of boys to girls in 6:5 is

Sol: Total = 48

Boys : Girls = 5 : 3

No. of Boys = $\frac{5}{\cancel{8}} \times \cancel{48}^{6} = 30$

No. of Girls = $48 - 30 = 18$

Girls added = G

$\frac{\text{Boys}}{\text{Girls}} = \frac{30}{18+G} = \frac{6}{5}$

$\cancel{30}^{5} \times 5 = \cancel{6}(18+G)$

$25 - 18 = G$

$G = 7$

LESSON-6: INCOME EXPENDITURE

Income = Expenditure + Savings

Q) The ratio of income of P and Q is 3:4 and the ratio of their expenditure is 2:3. If both of them save Rs 6000 each, the income of P is

Take any only one in terms of x.

Sol: Income $\Rightarrow$ P : Q = 3 : 4 → 3x, 4x

Expen $\Rightarrow$ P : Q = 2 : 3

Both save 6000

Income − Savings = Exp

$\frac{3x - 6000}{4x - 6000} = \frac{2}{3}$ $\Rightarrow 9x - 18000 = 8x - 12000$

$x = 6000$

Income of P = $3x = 3(6000)$

$= 18000$

Q) A and B have monthly incomes in the ratio 5:6 and monthly expenditures in the ratio 3:4. If they save ₹1800 and ₹1600, find the monthly income of B.

Sol: Income → A : B = 5 : 6 Expen → A : B = 3 : 4

A saving = 1800

B Saving = 1600

monthly income of B = ?

Let income of A = $5x$ and B = $6x$

$\because$ Income - Savings = Exp

$$\frac{5x - 1800}{6x - 1600} = \frac{3}{4}$$

$$20x - 7200 = 18x - 4800$$

$$2x = 2400$$

$$x = 1200$$

Income of B = $6x$ = $6(1200)$ = 7200.

Q) A man spends a part of his monthly income and saves a part of it. The ratio of his expenditure to his saving is 26:3. If his monthly income is ₹7250, what is the amount of his monthly savings?

Sol: $\frac{\text{Exp}}{\text{Saving}} = \frac{26}{3}$

29)7250(25
58
145
145

Income = 7250 savings = ?

Income = Exp + Saving

$7250 = 26x + 3x$

$\cancel{29}x = \cancel{7250}$ 250

$x = 250$

Savings = $3x$ = $3(250)$

= 750

Q) The monthly salary of A, B, C is in the proportion of 2:3:5. If C's monthly salary is Rs. 12000 more than that of A, then B's annual salary is :

Sol: Income monthly → A : B : C = 2 : 3 : 5

C's monthly salary = 12000 + A

$C - A = 12000$

$5x - 2x = 12000$

$3x = 12000$

$x = 4000$

B's monthly salary = $3x$ = 12000

B's annual salary = 12 (12000)

= 1,44,000

Q) The incomes of A and B are in the ratio 5:3. The expenses of A, B and C are in the ratio 8:5:2. If C spends Rs. 2000 and B saves Rs. 700 then A saves

Sol: Income → A : B = 5 : 3
$5y \quad 3y$

exp → A : B : C = 8 : 5 : 2
$8x \quad 5x \quad 2x$

exp of C = 2000 exp of A = $8x$ = 8000 exp of B = $5x$ = 5000

$2x = 2000$

$x = 1000$

B saves = 700

Income of B = Saves + exp

$3y = 700 + 5000$

$3y = 5700$

$y = 1900$

Income of A = $5y$ = 5(1900) = 9500

A Saves = Income - Exp

= 9500 - 8000

= 1500

LESSON-7: COINS

9) In a bag, there are three types of coins - 1 rupee, 50 paise and 25 paise in the ratio 3:8:20. Their total value is Rs. 372. Find the total number of coins?

Sol: Types of coins in bag - 1 rupee, 50 paise, 25 paise

ratio of coins = 3 : 8 : 20

Total value = Rs. 372

Total 1 rupee coins = $3x$
50 paise coins = $8x$
25 paise coins = $20x$

Total coins = ~~$3x+8x+20x$~~
= ~~$31x$~~

$\because$ 1 rupee = 100 paise
50 paise = $\frac{1}{2}$ rupee
25 paise = $\frac{1}{4}$ rupee

Total value = $3x$ rupees + $8x\left(\frac{1}{2}\right)$ rupee + $20x\left(\frac{1}{4}\right)$ rupee = Rs 372

$= 3x + \cancel{8}^{4}x\left(\frac{1}{\cancel{2}}\right) + \cancel{20}^{5}x\left(\frac{1}{\cancel{4}}\right) = 372$

$12x = 372$

$x = 31$

Total number of coins $= 3x + 8x + 20x$
$= 31x$
$= 31(31)$
$= 961$

Q) In a bag, there are three types of coins - 1 rupee, 50 paise and 25 paise in the ratio 2:3:4. Their total Value is Rs. 180. Find the total number of 50 paise coins?

Sol: coins ratio = 2:3:4

No. of 1 rupee coins = $2x$

No. of 50 paise coins = $3x$

No. of 25 paise coins = $4x$

Total value = Rs 180

Rs 180 = $2x$ Rupees + $3x\left(\frac{1}{2}\right)$ rupees + $4x\left(\frac{1}{4}\right)$ rupees

$$180 = \frac{4x + 3x + 2x}{2}$$

$$\overset{20}{\cancel{180}} \times 2 = \cancel{9}x$$

$$x = 40$$

No. of 50 paise coins = $3x = 3(40) = 120$

Q) In a bag, there are three types of coins - 1 rupee, 50 paise and 25 paise in the ratio 8:5:3. Their total value is Rs. 225. Find the total number of 1 rupee coins?

Sol: Coins ratio = 8:5:3

No. of 1 rupee coins = $8x$

No. of 50 paise coins = $5x$

No. of 25 paise coins = $3x$

Total value = Rs 225

Rs 225 = $8x$ rupees + $5x\left(\frac{1}{2}\right)$ rupees + $3x\left(\frac{1}{4}\right)$ rupees

$$225 = \frac{32x + 10x + 3x}{4}$$

$$\underset{5}{\cancel{225}} \times 4 = \cancel{45}x$$

$$x = 20$$

No. of 1 rupee coins = $8x = 8(20) = 160$

9) A box contains 420 coins of 1 rupee, 50 paise and 20 paise coins. The ratio of their values is 13 : 11 : 7. The number of 50 paise coins is?

Sol: Total No. of coins = 420

Ratio of Values = 13 : 11 : 7 $\longrightarrow$ convert this ratio in terms of No. of coins

~~Total value of 1 rupee = 13x~~

~~Total value of 50 paise = 11x~~

~~Total value of 20 paise = 7x~~

1 Rs $\longrightarrow$ 100 p $\longrightarrow$ 1 coins

1 Rs $\longrightarrow$ 20 p $\longrightarrow$? 5c

~~Rs~~ 13 : 11 : 7

1 rupee $\downarrow$ we will have 13 one Rupee coins

50 paise $\downarrow$ Two 50 p coins = 1 Rs

1 Rs $\rightarrow$ Two 50 p

11 Rs $\rightarrow$?

= 11 × 2

= 22 coins

20 paise $\downarrow$ Five 20 p coins = 1 Rs

1 Rs $\longrightarrow$ 5 20 p coins

7 Rs $\longrightarrow$?

= 7 × 5

= 35 coins

∴ No. of coins ratio = 13 : 22 : 35

420 coins = $13x + 22x + 35x$

$\cancel{420} = \cancel{70}x$

$x = 6$

No. of 50 p coins = $22x$

$= 22 \times 6$

$= 132$

9) A box contains 280 coins of 1 rupee, 50 paise and 25 paise coins. The ratio of their values is 8 : 4 : 3. The number of 50 paise coins is?

Sol: Total coins = 280

Ratio of values = 8 : 4 : 3

↓

Convert value ratio into no. of coins ratio

8 : 4 : 3

↓ 1 Rs values ↓ 8 Rupee coins

↓ 50 p values ↓ 1Rs = 2 50p coins ↓ 4Rs = ? = 4×2 = 8 coins

↓ 25 p values ↓ 1Rs = 4 25p coins ↓ 3 Rs = ? = 3×4 = 12 coins

Coins ratio = 8 : 8 : 12

$280 = 8x + 8x + 12x$

$280 = 28x$

$x = 10$

No. of 50p coins = $8x = 8(10)$

$= 80$ //

LESSON - 8 : AGES

Q) The ratio of the ages of two student in 3:2 one is elder than the other by 5 years what is the age of the younger student.

Sol: $S_1 : S_2 = 3 : 2$

$S_1 \sim S_2 = 5$ years

$3x \sim 2x = 5$ years

$x = 5$ years

Age of younger $= 2x = 2(5) = 10$ years

Q) The ratio of age of two boys is 5:6. After two years, the ratio will be 7:8. What will be the ratio of their age after 12 years?

Sol: $b_1 : b_2 = 5 : 6$

After 2 years:

$b_1 = (5x + 2)$ years $\qquad b_2 = (6x + 2)$ years

$$\frac{5x+2}{6x+2} = \frac{7}{8} \implies 40x + 16 = 42x + 14$$

$2x = 2$

$x = 1$

After 12 years:

$b_1 = 5x + 12$

$= 5(1) + 12$

$= 17$

$b_2 = 6x + 12$

$= 6(1) + 12$

$= 18$

$$\frac{b_1}{b_2} = \frac{17}{18}$$

Q) Harsha is 40 years old and Ritu is 60 years old. How many years ago was the ratio of their ages 3:5?

Sol: Harsha = 40 years
Ritu = 60 years } Present ages

Let years ago be x

$$\frac{\text{Harsha}}{\text{Ritu}} = \frac{40-x}{60-x} = \frac{3}{5}$$

$$200 - 5x = 180 - 3x$$

$$20 = 2x$$

$$x = 10 \text{ yrs}$$

Q) The ratio of the present ages of two brothers is 1:2 and 5 years back the ratio was 1:3. What will be the ratio of their ages after 5 years?

Sol: Present age ratio $b_1 : b_2 = 1:2$

<u>5 years back</u>:

$b_1 = x - 5$ $b_2 = 2x - 5$

$$\frac{b_1}{b_2} = \frac{x-5}{2x-5} = \frac{1}{3}$$

$$3x - 15 = 2x - 5$$

$$x = 10$$

<u>After 5 years</u>

$b_1 = x + 5$ $b_2 = 2x + 5$

$$\frac{b_1}{b_2} = \frac{x+5}{2x+5} = \frac{10+5}{2(10)+5} = \frac{\cancel{15}^{3}}{\cancel{25}_{5}} = \frac{3}{5}$$

LESSON-9 : MIXTURE

Q) A mixture contains alcohol and water in the ratio 4:3. If 5 litres of water is added to the mixture, the ratio becomes 4:5. Find the quantity of alcohol in the given mixture.

Sol: Initial alcohol : water $= 4:3$

water 5 lts added:

$$\text{alcohol} = 4x \cancel{+5}$$
$$\text{water} = 3x+5$$

$$\frac{\text{alcohol}}{\text{water}} = \frac{4x\cancel{+5}}{3x+5} = \frac{4}{5}$$

$$\cancel{20x + 25 = 12x + 20}$$

$$\cancel{8x}$$

$$5(4x) = 4(3x+5)$$
$$20x = 12x + 20$$
$$\cancel{8}_2 x = \cancel{20}_5$$
$$x = \frac{5}{2}$$

$$\text{alcohol} = 4x = \cancel{4}^2 \times \frac{5}{\cancel{2}} = 10 \text{ lts}$$

Q) In two alloys A and B, the ratio of zinc to tin is 5:2 and 3:4 respectively. Seven kg of the alloy A and 21 kg of the alloy B are mixed together to form a new alloy. What will be the ratio of zinc and tin in the new alloy?

Sol:

Alloy A

zinc : tin $= 5:2$

~~Total weight of Alloy A~~ $= \cancel{5x+2x}$
$\cancel{= 7x}$

Alloy B

zinc : tin $= 3:4$

~~Total weight of Alloy B~~ $= \cancel{3y+4y} \cancel{= 7y}$

A	B	C
$Z:t = 5:2$	$Z:t = 3:4$	7 kg of A + 21 kg B
$Z = \frac{5}{5+2}$ $t = \frac{2}{5+2}$	$Z = \frac{3}{3+4}$ $t = \frac{4}{3+4}$	$Z:t = (5+9):(2+12)$
$Z = \frac{5}{7}$ $t = \frac{2}{7}$	$Z = \frac{3}{7}$ $t = \frac{4}{7}$	$= 14:14$
7 kg of Alloy A :	21 kg of alloy B :	$= 1:1$
$Z = \frac{5}{7} \times 7$ $t = \frac{2}{7} \times 7$	$Z = \frac{3}{7} \times 21$ $t = \frac{4}{7} \times 21$	
$Z = 5$ $t = 2$	$Z = 9$ $t = 12$	

Q) Zinc and copper are in the ratio 5:3 in 400 gm of an alloy. How much of copper (in grams) should be added to make the ratio 5:4?

Sol: $Z : C = 5:3$

400 gm of alloy:

$5x + 3x = 400 \Rightarrow 8x = 400 \Rightarrow x = 50$

weight of Z $= 5x = 5(50) = 250$ gm

weight of C $= 3x = 3(50) = 150$ gm

New ratio $Z:C = 5:4$

Let copper added $= y$

$$\frac{Z}{C} = \frac{250}{150+y} = \frac{5}{4}$$

$1000 = 750 + 5y$

$5y = 250$

$y = 50$ gm

9) In a mixture of three varieties of tea, the ratio of their weights is 4:5:8. If 5 kg tea of the first variety, 10 kg tea of the second variety and some quantity of tea of the third variety are added to the mixture, the ratio the weights of three varieties of tea becomes as 5:7:9. In the final mixture, the quantity (in kg) of the third variety of tea was:

Sol: $T_1 : T_2 : T_3 = 4 : 5 : 8$

~~$T_1 = \frac{4}{17}$~~ ~~$T_2 = \frac{5}{17}$~~ ~~$T_3 = \frac{8}{17}$~~

~~5kg of T_1~~ $\downarrow$ ~~$\frac{4}{17} \times 5 = \frac{20}{17}$~~

~~10kg of T_2~~ $\downarrow$ ~~$\frac{5}{17} \times 10 = \frac{50}{17}$~~

~~x kg of T_3~~ $\downarrow$ ~~$\frac{8}{17} \times x = \frac{8x}{17}$~~

~~New ratio = 5:7:9~~

i.e. ~~$\frac{20}{17} : \frac{50}{17} : \frac{8x}{17} = 5:7:9$~~

~~$\frac{50}{17} : \frac{8x}{17} = 7:9$~~

$T_1 : T_2 : T_3 = 4 : 5 : 8$
$\quad\quad\quad\quad\quad\quad \downarrow \quad \downarrow \quad \downarrow$
$\quad\quad\quad\quad\quad\quad 4x \quad 5x \quad 8x$

New mixture: $(4x+5) : (5x+10) : (8x+t) = 5:7:9$

$(4x+5) : (5x+10) = 5:7$

$\frac{4x+5}{5x+10} = \frac{5}{7} \Rightarrow 28x + 35 = 25x + 50$

$3x = 15$

$\boxed{x = 5}$

$(5x+10) : (8x+t) = 7:9$

$\frac{25+10}{40+t} = \frac{7}{9}$

$\frac{\cancel{35}^{5} \times 9}{\cancel{7}} = 40+t \Rightarrow t = 45-40$

$t = 5$

$\therefore$ In New mixture, the quantity of third variety $= 8x + t$

$$= 8(5) + 5$$

$$= 45 \text{ kg}$$